

Coverage Control for a Group of UAVs and UGVs with the Effect of UAVs Altitude

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Abstract—This paper investigates coverage control for a group of aerial and ground robots. Our analysis reveals a novel cost function for placing the UAVs at the optimal height by considering a realistic model for the robots. This function provides a trade-off between the increasing coverage and reduction in quality of sensors which leads to a gradient-based distributed controller. This controller makes the robots move toward the optimal positions where they encode optimal coverage and minimizes the overlap of the sensor field of view (FoV). Local optimal coverage is shown using the Lyapunov stability theorem. Simulation results are also provided to demonstrate the effectiveness of the deployment schemes.

Index Terms—Multi-agent systems, Coverage control, Voronoi diagram, Quadrotor helicopter, UAVs, UGVs

I. INTRODUCTION

A multi-agent system includes several agents interacting with each other to accomplish a unique goal. Cooperative control of multi-agent systems has numerous applications, one of which is coverage control that allocates robots to provide optimal solutions for monitoring based on a defined objective function. Applications of coverage control includes border surveillance, identifying and exploring an unknown hazardous environment, and search and recovery operations [1]. In [2], a coverage control using the Voronoi algorithm is proposed. A similar method is used in [3] considering a real model for robots with limited sensing range. The author in [4] uses a distributed algorithm for coverage using robots with different dynamics and capabilities. Using the multiplicatively weighted Voronoi diagram, the robots minimize the introduced cost function. Coverage in an unknown environment with barriers has been investigated in [5]. The robots avoid collisions with existing obstacles by a proposed virtual density function. In [6], two different robots, aerial and ground ones, cover an unknown area. The ground robots are serving agents responsible for coverage, and the aerial ones are explorer robots that identify, explore, and detect the targets. [7] and [8] utilize aerial robots equipped with cameras to monitor a two-dimensional area. In [9], a multi-UAV system address coverage in precision agriculture. The same problem is investigated in [10], where the authors apply Deep Learning to control the UAVs. The author in [11] covers an environment by a group of quadrotors using locational optimization techniques. However, it does not addresses the effect of UAVs' altitude in coverage.

Deployment of multiple UAVs is of great importance in wireless networks. In this field, UAVs act as aerial base stations which feature desirable characteristics for rapid, on-demand, or emergency communications. Authors in [12] address the UAV deployment problem by analyzing the trade-offs between crucial system parameters such as height, antenna beam-width, and number of UAVs. In [13] and [14], UAVs emerge with the internet of things (IoT) technology and wireless sensor networks to collect data from devices that have limited transmit power.

In this article, a heterogeneous multi-agent system is introduced in which robots have different nonholonomic dynamics. Several previous articles considered cooperation between aerial and ground robots. The collaboration of a UAV and a UGV, which have comparable motion patterns, but different velocity constraints, is addressed in [15]. The authors propose a novel moving center trajectory for the UAV with the help of the UGV, which determines the center of trajectory. In [16], a robotic system consisting of an aerial and ground robot is investigated. The UAV acts as a leader, and the UGV follows it to accomplish the search operations. In [17] and [18], the UAVs are equipped with cameras that fly above the UGVs to inform the position, obstacles, and any possible danger. In [19], the collaboration of UAVs and UGVs is applied for information gathering and control in urban surveillance.

In this article, the aerial robot is a quadrotor helicopter, while the wheeled robot has second-order differential dynamics with omnidirectional sensors. With the mixture of these two types of robots, we can take advantage of the aerial robot, which has a higher speed and sensing range and the higher accuracy of the ground robot to optimize the coverage. As the aerial robot moves in three dimensions, the change in its altitude causes different FoV. This fact can play an essential role in the quality and scale of coverage. Section II reviews the mathematical definition of the coverage problem and the solution, which is based on the centroidal Voronoi partitions. Section III provides the robots' dynamics and their controllers. The coverage is investigated in Section IV. A novel specific cost function is derived in section V to find the optimal altitude for each UAV. The effectiveness of the proposed method is then evaluated through numerical simulations shown in sub-section D.

II. COVERAGE CONTROL PROBLEM

A. Problem Statement

To achieve optimum coverage, the problem is defined in mathematical models. In [2], the coverage is modeled as a locational optimization function. Then, the controllers are calculated to minimize the function. Using [2], we define the cost function as (1).

$$H(P_i) = \sum_{i=1}^n \int_{V_i} f(d)\phi(q)dq \quad (1)$$

d is the distance between point (q) and the position of sensor i (P_i). $f(d)$ is Euclidean distance, with $\phi(q)$ being the density function of the environment representing the importance of each point. The Centroidal Voronoi diagram is used in which a Voronoi cell is assigned to each robot, and they are steered toward their corresponding center of gravity [2].

III. ROBOT MODELS AND THE CONTROLLERS

A. The Aerial Robot

The aerial robot dynamics is shown in (2), [20].

$$\begin{aligned} \ddot{\phi} &= \left(\frac{I_y - I_z}{I_x}\right)\dot{\theta}\dot{\psi} - k_{fax}\dot{\phi}^2 - J_r\Omega_r\dot{\phi} + \frac{L}{I_x}U_2 \\ \ddot{\theta} &= \left(\frac{I_z - I_x}{I_y}\right)\dot{\phi}\dot{\psi} - k_{fay}\dot{\theta}^2 - J_r\Omega_r\dot{\theta} + \frac{L}{I_y}U_3 \\ \ddot{\psi} &= \left(\frac{I_x - I_y}{I_z}\right)\dot{\phi}\dot{\theta} - k_{faz}\dot{\psi}^2 + \frac{k_d}{I_z}U_4 \\ \ddot{x} &= \frac{1}{m}\{(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)U_1 - k_{ftx}\dot{x}\} \\ \ddot{y} &= \frac{1}{m}\{(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)U_1 - k_{fty}\dot{y}\} \\ \ddot{z} &= \frac{1}{m}\{(\cos\phi\cos\theta)U_1 - k_{ftz}\dot{z}\} - g \end{aligned} \quad (2)$$

By defining the state variables x_1 to x_{12} , the state equation of the robot dynamics is written as follows.

$$\dot{X} = \begin{bmatrix} x_2 \\ a_1x_6x_4 - K_{fax}x_2^2 - J_r\Omega_rX_4 + b_1U_2 \\ x_4 \\ a_2x_2x_6 - K_{fay}x_4^2 - J_r\Omega_rX_2 + b_2U_3x_6 \\ a_3x_2x_4 - K_{faz}x_6 + b_3U_4 \\ x_8 \\ \frac{1}{m}\{(\cos\phi\cos\theta)U_1 - k_{ftz}x_8\} - g \\ x_{10} \\ \frac{1}{m}\{(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)U_1 - k_{ftx}x_{10}\} \\ x_{11} \\ \frac{1}{m}\{(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)U_1 - k_{fty}x_{12}\} \end{bmatrix} \quad (3)$$

where $x_1 = \phi$, $x_2 = \dot{\phi}$, $x_3 = \theta$, $x_4 = \dot{\theta}$, $x_5 = \psi$, $x_6 = \dot{\psi}$, $x_7 = z$, $x_8 = \dot{z}$, $x_9 = x$, $x_{10} = \dot{x}$, $x_{11} = y$, $x_{12} = \dot{y}$

$$\begin{aligned} a_1 &= \left(\frac{I_y - I_z}{I_x}\right) b_1 = \left(\frac{L}{I_x}\right) a_2 = \left(\frac{I_z - I_x}{I_y}\right) \\ b_2 &= \left(\frac{L}{I_y}\right) a_3 = \left(\frac{I_x - I_y}{I_z}\right) b_3 = \left(\frac{K_d}{I_z}\right) \end{aligned}$$

TABLE I
DEFINITION OF SYMBOLS

symbols	definition	symbols	definition
x, y, z	the position of the quadrotor	$I_{x,y,z}$	body inertia
ϕ, θ, Ψ	roll, pitch and yaw angle	J_r	rotor inertia
$k_{fax}, k_{fay}, k_{faz}$	fraction coefficient	Ω_r	rotor speed
U_1, U_2, U_3, U_4	controllers input	m, L	Mass, lever
g	gravity acceleration	k_d	drag coefficient

B. The Control Strategy of the Aerial Robot

The back-stepping approach proposed in [21] is utilized.

C. The Control of the UAV

According to (3), the subsystems of angles and altitude (z) are in the appropriate formats to use the back-stepping method. Thus, the following controllers are introduced [21].

$$\begin{aligned} U_2 &= \frac{1}{b_1}(z_1 - c_2z_2 - a_1x_4x_6 + a_2x_4\Omega_r + \ddot{x}_{1d} - c_1\dot{z}_1) \\ U_3 &= \frac{1}{b_2}(z_3 - c_4z_4 - a_3x_2x_6 + a_4x_4\Omega_r + \ddot{x}_{3d} - c_3\dot{z}_3) \\ U_4 &= \frac{1}{b_3}(z_5 - c_6z_6 - a_5x_2x_4 + \ddot{x}_{5d} - c_5\dot{z}_5) \\ U_1 &= \frac{m}{\cos\phi\cos\theta}(Z_7 - z_8 + k_{ftz}x_8 + g) \end{aligned} \quad (4)$$

where x_{id} is the desired value of the state x_i , with z_i being its corresponding error. By assuming small values for roll and pitch and considering zero as the desired value for yaw, the equations of the position are simplified to the following form.

$$\begin{aligned} \ddot{x} &= \frac{U_1}{m}\theta_d - k_{fty}\dot{x} \\ \ddot{y} &= -\frac{U_1}{m}\phi_d - k_{fty}\dot{y} \end{aligned} \quad (5)$$

By selecting the angle of θ_d and ϕ_d as virtual controllers, the position can be controlled by a PID controller as follows.

$$\begin{aligned} U_x &= \frac{m}{U_1}(k_p(x_d - x) + k_d(\dot{x}_d - \dot{x}) + k_i \int (x_d - x)) \\ U_y &= -\frac{m}{U_1}(k_p(y_d - y) + k_d(\dot{y}_d - \dot{y}) + k_i \int (y_d - y)) \end{aligned} \quad (6)$$

where x_d and y_d are the desired position of the robot.

D. Ground Robot and its Controller

The UGV is a two-wheeled differential robot. The kinematic and dynamic equations of the robot are as follows [22]:

$$\begin{aligned} \dot{x} &= v\cos\theta \\ \dot{y} &= v\sin\theta \\ \dot{\theta} &= \omega \end{aligned} \quad (7)$$

(x, y) is the position of robots, θ is the angle between robot and x -axis, with v and ω being the speed and the angular acceleration. The controller is introduced in (8) using [22].

$$\begin{aligned} v &= -k_v e_x \\ \omega &= k_\omega \arctan\left(\frac{e_x}{e_y}\right) \end{aligned} \quad (8)$$

where K_v and K_ω are positive gains and e_x and e_y are errors with respect to the desired position.

IV. COVERAGE OF HETEROGENEOUS ROBOTS

In section A, it is assumed that the height of quadrotors does not change. Using the Voronoi diagram introduced in [2], the positions of both types of robots are controlled to move toward the center of their corresponding Voronoi partitions.

A. Coverage by wheeled robots and quadrotors

It is assumed that n_1 UGVs and n_2 UAVs are randomly spread in an area. The aim is to find a distributed controller so that the coverage will be done most effectively. The control law for i_{th} wheeled robots and j_{th} quadrotors are selected based on (8) and (4-6), respectively, with the desired position being the central point of their Voronoi partitions as follows.

$$\begin{aligned} U_{x_i} &= \frac{m}{U_1} (M_{vi}(C_{viX} - x_i) - k_x \dot{x}_i) \\ U_{y_i} &= \frac{m}{U_1} (M_{vi}(C_{viY} - y_i) - k_y \dot{y}_i) \\ \begin{bmatrix} e_{x_i} \\ e_{y_i} \end{bmatrix} &= \begin{bmatrix} \cos\theta_i & \sin\theta_i \\ -\sin\theta_i & \cos\theta_i \end{bmatrix} (p_i - C_{vi})^T \end{aligned} \quad (9)$$

Theorem: By selecting the controllers (9), the multi-agent system described above is asymptotically stable, and robots' positions converge to the locally optimum coordinates.

Proof: The following Lyapunov function is considered.

$$V = \frac{1}{2} (\sum_{i=1}^{n_1} H_i + \sum_{j=1}^{n_2} H_j + \sum_{j=1}^{n_2} \dot{p}_j^T \dot{p}_j) \quad (10)$$

where H_i is the cost function of agent i defining as (11) [2].

$$H_i = \int_{v_i} \|p_i - q\|^2 \varphi(q) dq \quad (11)$$

The derivative of the Lyapunov function with respect to the angles and the height is zero. Thus, we will have

$$\begin{aligned} \dot{V} &= \frac{1}{2} (\sum_{i=1}^{n_1} \frac{\partial H_i}{\partial p_i} \dot{p}_i + \sum_{j=1}^{n_2} \frac{\partial H_j}{\partial p_j} \dot{p}_j + \sum_{j=1}^{n_2} \dot{p}_j^T \ddot{p}_j) \\ &= \sum_{i=1}^{n_1} \dot{p}_i M_{vi} (P_i - C_{vi}) + \sum_{j=1}^{n_2} \ddot{p}_j (M_{vj} (P_j - C_{vj}) + \dot{p}_j) \end{aligned} \quad (12)$$

Where M and C are defined by interpreting the Voronoi partition as a physical object as follows [2].

$$M_{vi} = \int_{v_i} \varphi(q) dq, C_{vi} = \frac{\int_{v_i} q \varphi(q) dq}{M_{vi}}$$

By selecting the controllers (9) and replacing them in (8), the following expression is obtained:

$$\dot{V} = -k_{vi} M_{vi} \sum_{i=1}^{n_1} \|p_i - C_{vi}\|^2 - \sum_{j=1}^{n_2} (k_1 \dot{x}_j^2 + k_2 \dot{y}_j^2) \quad (13)$$

K_1 , K_2 and K_{vi} are positive constants.

The derivative of the Lyapunov function (14) is negative semidefinite. Suppose M is the set of points where (14) equals zero. Since the area is convex, this set of points is inside Q . Thus, M is a positively invariant set for the closed-loop

system, and the LaSalle principle can be used to prove the stability.

$$\begin{aligned} \dot{V} = 0 &\Rightarrow \begin{cases} \dot{x} \equiv 0 \Rightarrow \ddot{x} \equiv 0 \\ \dot{y} \equiv 0 \Rightarrow \ddot{y} \equiv 0 \end{cases} \\ &\Rightarrow \begin{cases} \ddot{x}_j = \frac{U_1}{m} U_{xj} - k_{fax} \dot{x} = 0 \\ \ddot{y}_j = \frac{U_1}{m} U_{yj} - k_{fay} \dot{y} = 0 \end{cases} \\ &\Rightarrow \begin{cases} U_{xj} = 0 \Rightarrow x_j = C_{vjx} \\ U_{yj} = 0 \Rightarrow y_j = C_{vjy} \end{cases} \Rightarrow p_j = C_{vj} \\ \dot{V} = 0 &\Rightarrow \|P_i - C_{vi}\|^2 = 0 \Rightarrow P_i = C_{vi} \end{aligned} \quad (14)$$

Therefore, the system is asymptotically stable, and the position of robots converges to their locally optimum coordinates.

B. Simulations

Four UAVs, each at a unique height, and four UGVs start moving from the lower-left corner. Fig. 1 and Fig. 2 show the path of the robots. The robots cover the area inside the convex area, with sensors being defined in a circle. Since robots must move only within their defined range, they are assumed to be fully aware of the geography of the environment. The graph

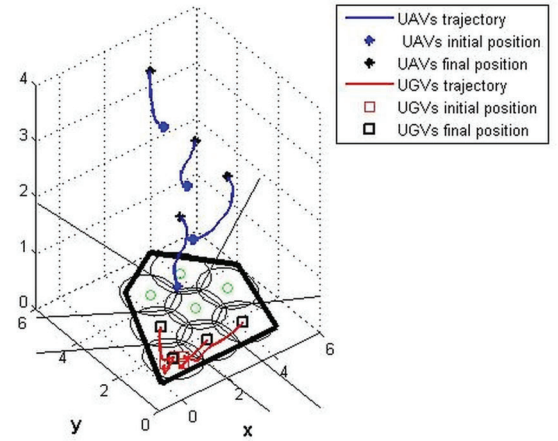


Fig. 1. simulation results by two UGVs and UAVs.

of the robots' position versus time is shown in Fig. 3 and Fig. 4. The positions of the robots remain constant after 30s. This indicates the stability of the proposed method.

The graph of UAVs' orientation and altitudes is illustrated in Fig. 5 to Fig. 8. Although roll and yaw are not controlled independently, they are stable and converge to zero. High peaks in curves are due to the high error in the first moment.

V. COVERAGE CONTROL OF UAVS BY CHANGING THEIR ALTITUDES

None of the previous research investigated the impact of changes in UAVs height. However, coverage zone (FoV) and the accuracy and performance of sensors alter according to

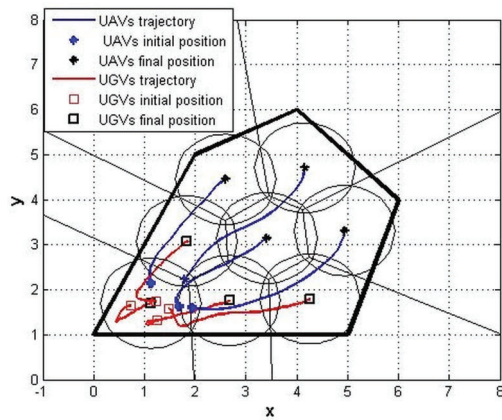


Fig. 2. the projection of movement in two dimensions.

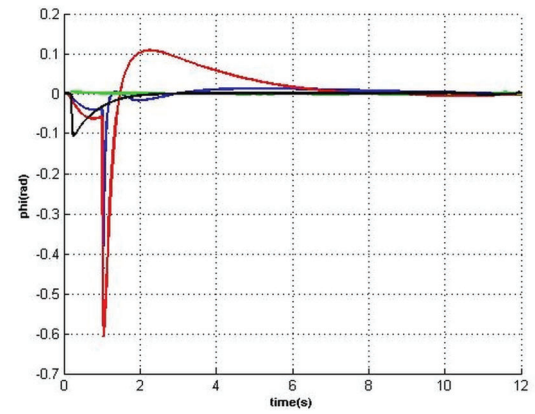


Fig. 5. The Pitch of UAVs VS time.

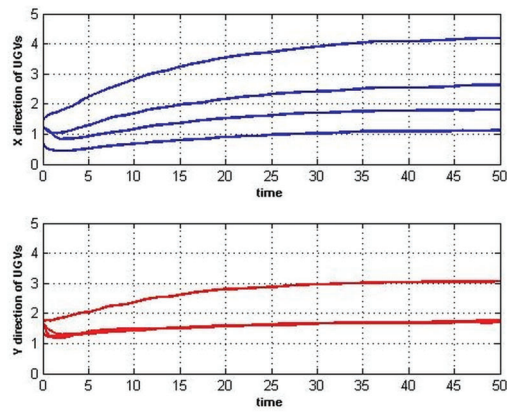


Fig. 3. the position of UGVs VS time.

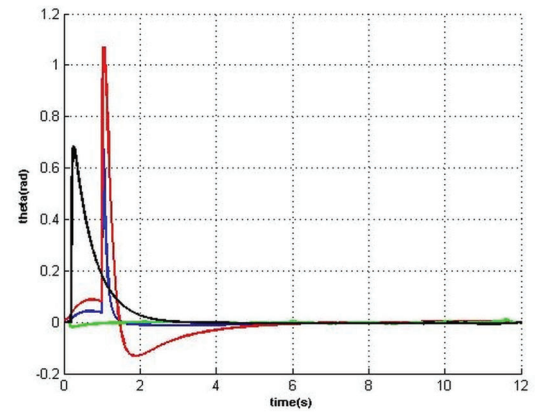


Fig. 6. The roll of UAVs VS time.

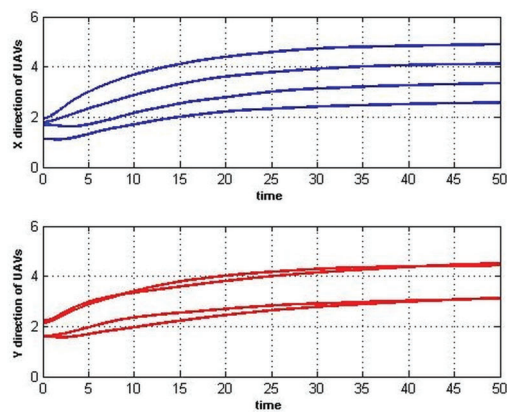


Fig. 4. The position of UAVs VS time.

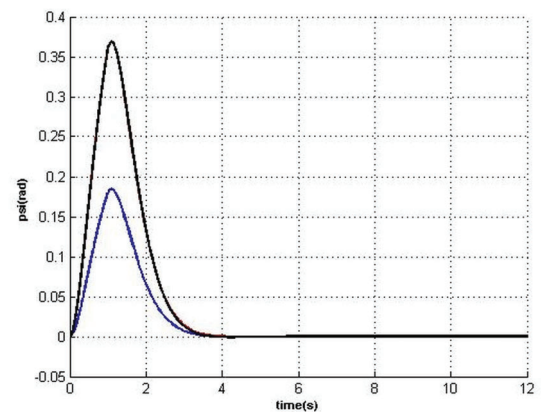


Fig. 7. The yaw of UAVs VS time.

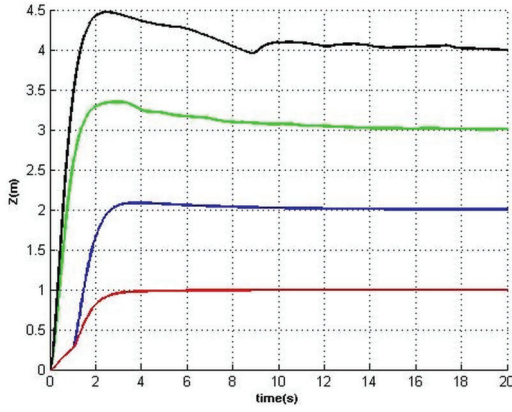


Fig. 8. The altitude of UAVs VS time.

the distance of UAVs from the ground. The relation between height and FoV is depicted in Fig. 9. The sensor can be a high-speed camera for search missions or an antenna with an isotropic transmitter for radio coverage in wireless communications. The next section examines the impact of height change in coverage.

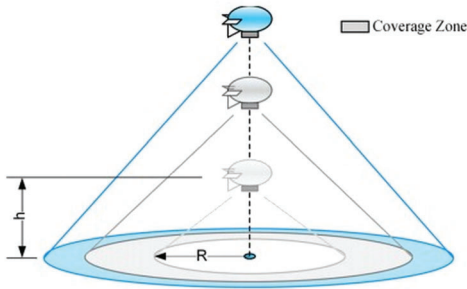


Fig. 9. Coverage zone or field of view of a UAV [23].

A. Modeling the height change in coverage

As shown in Fig. 9, the higher the quadcopter flies, the wider sensing range is achieved. On the other hand, increasing altitude will affect the quality of the coverage. The goal is to find the optimum height where the whole environment is covered, and in important points, the height is relatively lower and vice versa. The following cost function is proposed to achieve this goal.

$$H = \sum_{i=1}^N \int_{v_i} \frac{\|p_i - q\|^2}{z_i^2} \phi(q) + \int_{v_i} \phi(q)^2 z_i dq \quad (15)$$

In (15), initially, the value of H decreases as z increases. However, the function's output increases after a certain height

that depends on the density function, the Voronoi regions, and the robots' position.

The UAV sensor range is directly related to its height. For example, if the quadrotor is equipped with a camera and the angle of view of its lens is 45 degrees, then the range of the quadrotor sensor will be the same as its height [8]. Regarding UAVs in wireless communication, the coverage zone is considered a coverage disk whose radius is directly related to its altitude according to its beam-width. For other sensors, these two quantities can be related by a proportional coefficient. Owing to this fact, the height square in the denominator of the first term in H indicates that the cost decreases if the robot flies higher. The second term represents the volume of a cone with height z and base area. In areas where the density function is high, the height should be reduced to decrease the volume of the cone and thus the value of the cost function. According to this function, the control law for robots is determined and its performance is evaluated using numerical simulation.

B. The partial derivative of the cost function

The partial derivative of the cost function with respect to the position of the robot is as follows.

$$\begin{aligned} \frac{\partial H}{\partial p_i} &= 2 \int_{v_i} \frac{1}{z_i^2} (p_i - q) \phi(q) dq \\ &= \frac{2M_{v_i}}{z_i^2} (p_i - C_{v_i}) \end{aligned} \quad (16)$$

The partial derivative of H with respect to z will be.

$$\begin{aligned} \frac{\partial H}{\partial p_i} &= \sum_{j=1}^{N_a} \frac{\partial}{\partial z_i} \int_{v_j} \left(\frac{k}{z_j^2} \|p_j - q\|^2 + \phi(q) z_i \right) \phi(q) dq \\ &= \sum_{j=1}^{N_a} \int_{v_j} \phi(q) \frac{\partial}{\partial z_i} \left(k \frac{\|p_j - q\|^2}{z_j^2} + \phi(q) z_i \right) dq \\ &= -2 \int_{v_j} \phi(q) \frac{\|p_j - q\|^2}{z_j^3} dq + \int_{v_i} \phi(q)^2 dq \\ &= -\frac{2I_{v_i}}{z_i^3} + \int_{v_i} \phi(q)^2 dq \end{aligned} \quad (17)$$

The critical points with respect to the robot position are equal to the center of the Voronoi partition. Likewise, the second critical point will be

$$\begin{aligned} \frac{\partial H}{\partial z_i} &= -2 \frac{I_{v_i}}{z_i^3} + \int_{v_i} \phi(q)^2 dq = 0 \\ \Rightarrow z_i^3 &= \frac{2I_{v_i}}{\int_{v_i} \phi(q)^2 dq} \quad z_i = \left(\frac{2I_{v_i}}{M_{v_i}^2} \right)^{1/3} \end{aligned} \quad (18)$$

where $\int_{v_i} \phi(q)^2 dq$ is written as $M_{v_i}^2$.

Z is highly dependent on the density function. If $\phi(q)$ is high, the robot height will decrease to improve sensor accuracy; Likewise, if the priority is low, the height will increase and lead to a greater FoV, and consequently greater coverage zone.

C. Proposing the Control Law

As investigated, the control law should allow the robots to be directed to the local extreme point of H . According to (4-6) the following control law is proposed.

$$\begin{aligned} U_{1i} &= \frac{1}{\cos\phi\cos\theta} \left(-\frac{\partial H}{\partial z_i} - k_z \dot{z}_i + mg \right) \\ U_{x_i} &= k \frac{m}{U_{1i}} \left(-\frac{\partial H}{\partial x_i} - k_x \dot{x}_i \right) \\ U_{y_i} &= -k \frac{m}{U_{1i}} \left(-\frac{\partial H}{\partial y_i} - k_y \dot{y}_i \right) \end{aligned} \quad (19)$$

Theorem: A team of quadrotors with dynamics (2) and controllers (19) will provide optimum coverage which optimize the cost function (15).

Proof: The following Lyapunov function is considered.

$$V = \sum_{i=1}^{n_{uav}} \frac{1}{2} [2H_i + \dot{Z}_i^T \dot{z} + \dot{p}_i^T \dot{p}] \quad (20)$$

The derivative of the Lyapunov function is obtained as follows:

$$\begin{aligned} \dot{V} &= \sum_{i=1}^{n_{uav}} \dot{z}_i \dot{z}_i + \frac{\partial H}{\partial z} \dot{z}_i + \dot{p}_i^T \ddot{p}_i + \frac{\partial H}{\partial p} \dot{p}_i^T \\ &= \sum_{i=1}^{n_{uav}} \dot{z}_i \left(\dot{z}_i + \frac{\partial H}{\partial z} \right) + \dot{p}_i^T \left(\ddot{p}_i + \frac{\partial H}{\partial p} \right) \end{aligned} \quad (21)$$

By substituting the equivalent of $\ddot{p}_i$ and $\ddot{z}_i$ as written in (2) and using controller (20) we will have

$$\dot{V} = \sum_{i=1}^{n_{uav}} -(k_1 \dot{z}_i^2 + k_2 \dot{x}_i^2 + k_3 \dot{y}_i^2) \quad (22)$$

The derivative of the Lyapunov function is non-positive. Since the second derivative is bounded, the first derivative is uniformly continuous. Therefore, due to Barbalat's lemma [24], the derivative of Lyapunov's function converges to zero. Now, we find these corresponding points.

$$\begin{aligned} \dot{V} = 0 &\implies \dot{z}_i = 0 \implies \ddot{z}_i = 0 \\ \ddot{z}_i &= (\cos\phi\cos\theta)U_1 - k_{ftz}\dot{z}_i - mg = 0 \end{aligned} \quad (23)$$

using the control law written in the equation (4), we will have.

$$\begin{aligned} -\frac{\partial H}{\partial z_i} - k_z \dot{z}_i &= 0 \\ \implies \frac{\partial H}{\partial z_i} &= 0 \end{aligned} \quad (24)$$

Likewise, for the robots' position we will have

$$\begin{aligned} \dot{V} = 0 &\implies \dot{x}_i = 0 \implies \ddot{x}_i = 0 \\ \ddot{x}_i &= \frac{U_{1i}}{m} U_{x_i} - k_{fax} \dot{x}_i = 0 \\ U_{x_i} = 0 &\implies \frac{\partial H}{\partial x_i} = 0 \\ \dot{V} = 0 &\implies \dot{y}_i = 0 \implies \ddot{y}_i = 0 \\ \ddot{y}_i &= \frac{U_{1i}}{m} U_{y_i} - k_{fay} \dot{y}_i = 0 \\ U_{y_i} = 0 &\implies \frac{\partial H}{\partial y_i} = 0 \end{aligned} \quad (25)$$

TABLE II
VALUE OF PARAMETERS

parameter	value	parameter	value
L	0.23	$I_{x,y,z}$	$10^{-3} \times (7.5, 7.5, 15)$
k_d	1.02	$k_{fax}, k_{fay}, k_{faz}$	$10^{-4} \times (0.6, 0.6, 7.5)$
m	0.65	C_1, C_2, C_3, C_4	10.5, 2, 10, 2
J_{r4}	6.5×10^{-5}	C_5, C_6, C_7, C_8	2, 2, 3.5, 1.5
g	9.81	k_x, k_y, k_z	2, 2, 1.5

According to the equations (22-25), the points where $\dot{V} = 0$ correspond to those where $\frac{\partial H}{\partial z} + \frac{\partial H}{\partial y} + \frac{\partial H}{\partial x} \equiv 0$, which are the extreme points of the cost function. Thus, the positions of the aerial robots converge to the critical points of the cost function.

D. Numerical Simulations

Simulations are carried out in three different scenarios to illustrate the performance of the proposed controller. Constant values are provided in table 2.

In the first step, ten robots with a communication radius of 4 meters are randomly placed in a 1×1 -meter square. They must optimally cover a 10×10 -meter square. The density function of the environment is a Gaussian function shown in (26). At the point $[10 \ 10]$, the value of ϕ is maximized. Thus, in the simulation result, the number of robots should be more, and their height should be relatively less around this particular point.

$$\phi = 100e^{\frac{-norm([qx \ qy] - [10 \ 10])}{5}} \quad (26)$$

Moving from their initial position and identifying their respective Voronoi regions, and by changing their height, the robots cover the two-dimensional environment, which is illustrated by Fig. 10. Because of the density function, the UAVs concentrate more on the central points. Moreover, the altitudes are so that the overlap of UAVs' FoV is minimized, and the coverage is maximized. This is shown in Fig. 11.

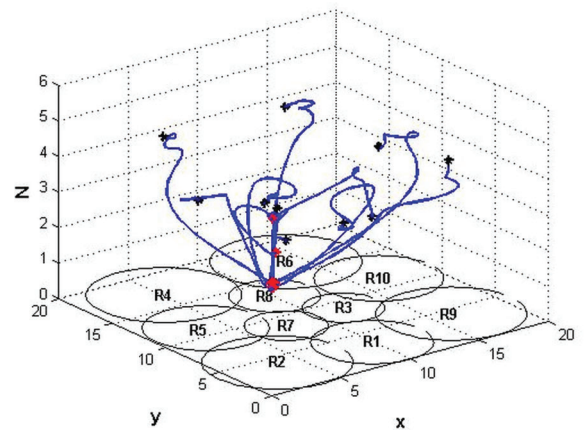


Fig. 10. The path of the robots from the initial position to the final position, the circles represent the field of view of robots.

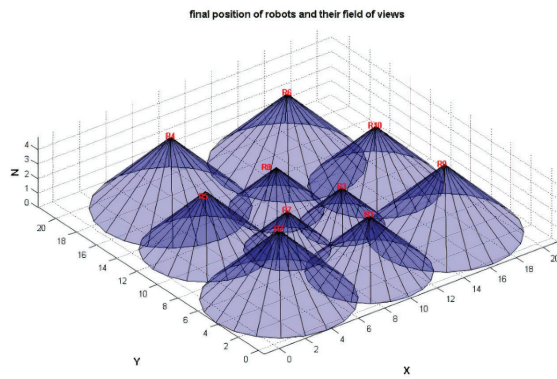


Fig. 11. The final position of the robots and their field of views.

In the second scenario, eight UAVs cover a convex area whose points have the same importance. Therefore, in the final stage, the robots should be all at the same height. Also, the height of each should be such that the least overlap and the highest coverage are achieved. The position and FoV of robots are shown in Fig. 12 and Fig. 13. Fig. 14 shows the position of the robots in terms of length, width, and height versus time. The positions of the robots are finally stable and the robots settle in their optimal positions.

Fig. 15 illustrates the coverage with four UAVs and five UGVs. The ground robots move within a two-dimensional environment, and the radius of their sensors is fixed (3m). In contrast, the UAVs maximize coverage by moving in a three-dimensional environment and changing their FoV. The robots start moving together and settle in their final arrangement to achieve the best coverage. Since there is no priority for ground or air robots, the quadrotors cover the farther points due to a higher speed, and UGVs are responsible for closer points. Fig. 16 and Fig. 17 illustrate the positions of robots versus time.

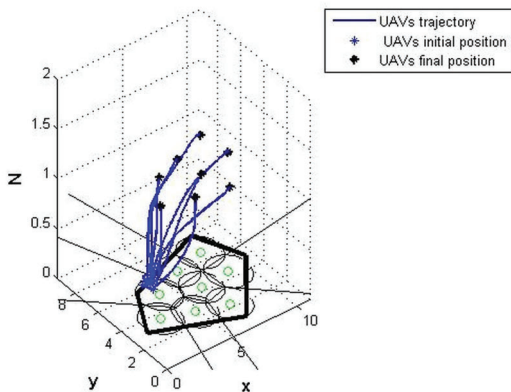


Fig. 12. The path of the robots from the initial position to the final position in the second scenario.

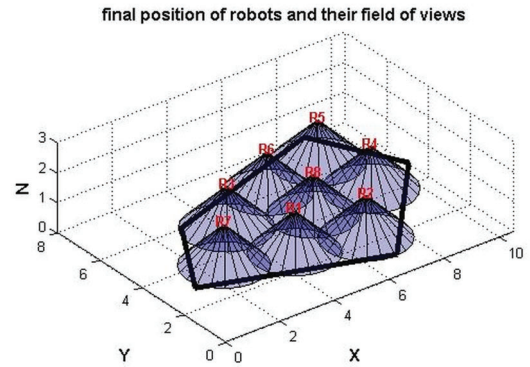


Fig. 13. The final position of the robots and their FoV, The density function of the environment is uniform, and the robots are almost at the same height.

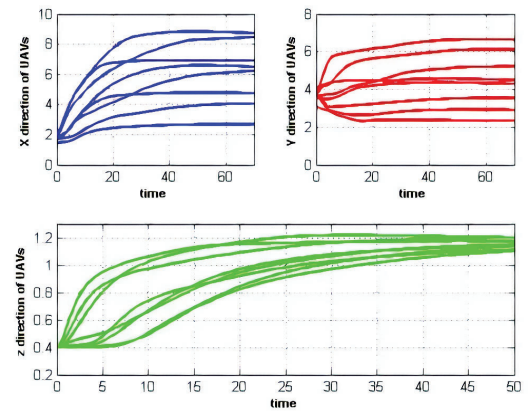


Fig. 14. The UAVs position versus time.

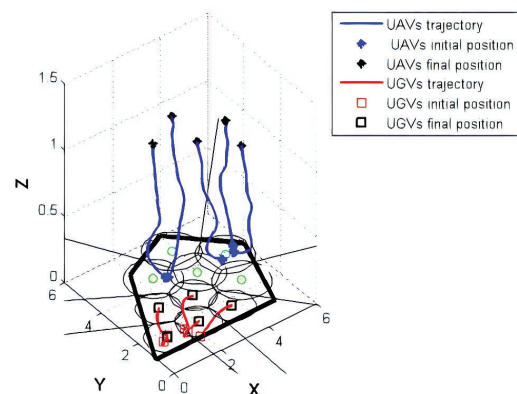


Fig. 15. The final position of the robots.

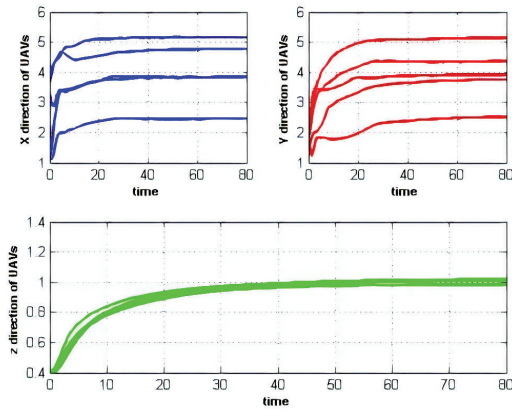


Fig. 16. The final position of the robots

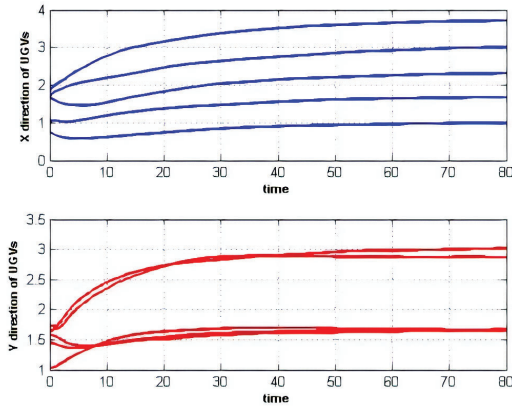


Fig. 17. The UGVs position versus time.

VI. CONCLUSION

In this study, we considered the coverage problem for a group of UAVs and UGVs by considering a more realistic model. We investigated the effect of altitude change for UAVs that led to an extension to the Voronoi-based standard coverage problem. The Lyapunov-based stability analysis showed that the vehicles finally converge to the centroidal Voronoi configuration, and the whole system is stable. The proposed approach has been successfully verified in numerical simulations.

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